# CHEN Liheng

## EDUCATION

**Hong Kong University of Science and Technology** **Aug. 2025 – Jun. 2029(expected)**

Doctor of Philosophy in Individualized Interdisciplinary Program (Computer Vision and Computer Graphics)

Supervisors: Hongbo Fu & Anyi Rao

**Beijing Normal University** **Sep. 2021 – Jun. 2025**

B.Eng. in Artificial Intelligence

Core Courses: Calculus (95), Fundamental Physics (90), Linux Operating System (93), Database and Data Principles (91)

Awards: First Prize Scholarship for the Competition, 2024 COMAP MCM F Prize

## PUBLICATION

**Li-Heng Chen** with NIO Foundation Model Research Team, served as **core-contributor.** *NWFM: A Vision-Native 4D World Foundation Model for Autonomous Driving.* **Technical Report (2026).**

**Li-Heng Chen**, Haokai Pang, Chengye Su, Jiarun Liu, Qifeng Chen, Ziqian Ni, Jianxin Huang, Shi-Sheng Huang, Hongbo Fu and Sheng Yang. *USR-Drive: Unified Driving Scene Representation via Joint Denoising of 3D Gaussians and Boxes.* **Under reviewed**.
arXiv: https://arxiv.org/abs/2608.19036

**Li-Heng Chen**, Ke Cheng, Yahui Liu, Lei Shi, Shi-Sheng Huang and Hongbo Fu (2026). *VistaGEN: Consistent Driving Video Generation with Fine-Grained Control Using Multiview Visual-Language Reasoning.* **Under reviewed**.
arXiv: https://arxiv.org/abs/2503.22349

**Li-Heng Chen**, Zi-Xin Zou, Chang Liu, Tianjiao Jing, Yan-Pei Cao, Shi-Sheng Huang, Hongbo Fu and Hua Huang (2025). *GCRayDiffusion: Pose-Free Surface Reconstruction via Geometric Consistent Ray Diffusion.* Accepted by **International Conference on Computer Vision 2025**. arXiv: https://arxiv.org/abs/2503.22349

Huang, S., Chen, G., **Chen, L.**, & Huang, H. (2024). *NeuralIndicator: Implicit Surface Reconstruction from Neural Indicator Priors.* Proceedings of the 41st **International Conference on Machine Learning**

## PROFESSIONAL EXPERIENCE

**NIO Autonomous Driving** **Mar. 2026 – Present**

***Algorithm Researcher Intern, Foundation Model Research Team***

- Designed a unified generative representation for dynamic driving scenes that encodes dense 3D Gaussian geometry and sparse 3D bounding boxes as aligned latent streams, using dual branch autoencoders, Unified Positional Encoding, and a Multi-Modal Diffusion Transformer for joint denoising, achieving 27.55 PSNR, 0.854 SSIM, and 0.076 LPIPS for reconstruction, together with 0.625 NDS and 0.552 mAP for 3D detection on nuScenes.
- Designed a geometry grounded 4D visual tokenizer that aggregates synchronized multiview driving driving videos through a DINOv2 initialized visual backbone, camera and time embeddings, frame attention, and global attention, with complementary depth, point map, Gaussian rendering, and bidirectional motion supervision, achieving 2.55m depth RMSE, 0.146m EPE3D, and 32.34 PSNR on Waymo.
- Designed a Perception transfer framework, adapting pretrained 4D visual tokens through lightweight interfaces for Gaussian

anchor based semantic occupancy prediction and StreamPETR style temporal 3D object detection, achieving 35.56 occupancy IoU and 21.50 semantic mIoU, together with 0.601 NDS and 0.543 mAP on nuScenes.
- Enabled a Video Generator by exposing the pretrained 4D tokenizer hierarchy as a geometry grounded generative state, allowing a Diffusion Transformer to predict future visual tokens that remain decodable into temporally coherent RGB videos and 3D geometry, achieving 298.72 FVD and 4.37m depth RMSE for 16 frame future prediction.

**Meituan Inc.** **Dec. 2025 – Mar. 2026**

***Algorithm Researcher Intern, Department of Unmanned Vehicle***

- Developed a controllable Multiview driving video generation framework based on 3DVAE and Spatial Temporal Diffusion Transformer, integrating BEV layouts, camera poses, text descriptions, reference images, and object trajectories to enable high resolution, long sequence, and fine-grained object level video generation.
- Designed a multimodal object conditioning and feature fusion module, combining Fourier encoded 3D geometric features, text embeddings, and SigLIP visual features with instance identity encoding to maintain object appearance across views and time, achieving 53.48 mAP, 65.70 IoU, 60.3% text alignment, and 70.7% image alignment.
- Built a generation, evaluation, and regeneration closed loop framework with a multiview Vision Language Model evaluator and 3D guided object refinement module to automatically detect and repair inconsistent objects, improving multiview and temporal consistency to 84.9% index consistency while supporting controllable generation of rare and long-tail driving objects.

**Vast.AI** **Aug. 2024 – Jun. 2025**

***Algorithm Researcher Intern, Department of Technology and Engineering***

- Developed a 3D reconstruction and camera pose estimation pipeline, using Dinov2 for accurate image feature extraction and 7D ray generation with Plücker coordinates, achieving over 90% accuracy for camera pose estimation on the Objaverse dataset
- Constructed a triplane network with MLP-based geometric feature analysis to generate Signed Distance Functions (SDF)
- Integrated SDF as a conditioning factor for the denoising process, using ray-calculated surface points as query points to significantly enhance the accuracy and fidelity of both 3D object shapes and camera pose estimation

**ABB (China)** **Jul. 2024**

***Algorithm Engineer Intern, Department of Digital Systems***

- Fine-tuned YOLOv8 parameters to develop a surveillance video integration program for detecting indoor population changes, and configured 5 modes to provide air conditioning strategies based on population density
- Developed interfaces and standardized the database format for integration with the company's industrial software Zee600
- Optimized data updates, real-time writing, and data structure for the smart screen system, eliminating 80% of the manual steps and enabling staff to access real-time data from four modules with a single IP address modification

**Xiamen Area of China (Fujian) Pilot Free Trade Zone Port Power Supply & Tech. Co., LTD.** **Jan. 2024 – Feb. 2024**

***System Engineering Intern, Department of Communications***

- Contributed to the usability optimization of the EAM system by resolving attribute conflicts in equipment inventory management and implementing automated equipment model selection, significantly reducing user workload
- Contributed to the blueprint design of the project management system for Harbour Electric. Designed the data structure for project information tables, laying the groundwork for subsequent development
- Engaged in testing projects for the Geographic Information System for the power information in the Harbour region, optimizing various system functionalities

## ACADEMIC EXPERIENCE

**The Power of Momentum: How to Win the Match with Data (2024 MCM Problem C)** **Feb. 2024**

***Team Leader***

- Developed a model based on IG-LSTM to measure player momentum, and conducted Correlation Analysis and Stochastic Simulation to assess the impact of momentum swings on match flow

- Utilized an information entropy-based decision tree to extract key features driving momentum swings, and designed an ANN-driven Momentum Swing Prediction Model (MSPM) to forecast match dynamics
- Optimized the model's accuracy and generalizability by incorporating new features, fine-tuning the parameters, and testing it across various tournaments and sports, securing the F prize

**Construction of a Speckle Pattern Model with Paddy Field Characteristics** **May 2022 – May 2023**

***Research Assistant*** **(under the supervision of Professor Cai Yongqiang)**

- Utilized OpenCV to perform grayscale conversion, smoothing filtering, and edge enhancement on images, followed by region segmentation and extraction of morphological and textural features
- Extracted object edges using Sobel and Canny algorithms, and enhance effectiveness through subsequent optimization using morphological operations, Hough transform, and other optimizations
- Conducted statistical analysis on image features and boundaries to derive characteristic parameters, and enhanced the conformity between models and actual field shapes by model optimization and reaction-diffusion equation optimization

## ADDITIONAL INFORMATION

**Languages**: Mandarin Chinese (native), English (proficient)

**Programming Skills**: C/C++, Python, SQL

**Hobbies**: Piano, Sports (basketball, swimming, cycling, fitness, etc.)